📄 Technical Report: Retail Sales Analysis

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1. Introduction

In today's competitive retail environment, data-driven decision-making plays a crucial role in improving business performance. This project conducts an end-to-end analysis of retail sales data to identify key revenue drivers, customer behaviour patterns, and operational inefficiencies.

The analysis integrates statistical methods, exploratory data analysis, and business intelligence techniques to generate actionable insights that can support strategic planning and operational improvements.

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2. Business Problem Statement

Retail organisations often face challenges such as:

- Uneven sales performance across regions
- Lack of understanding of customer behaviour
- Inefficient delivery systems
- Poor identification of high-value customers

This project aims to address these challenges by leveraging transactional data to uncover patterns and recommend data-driven strategies.

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3. Objectives

The primary objectives of this analysis are:

- Identify factors influencing revenue generation
- Analyse customer demographics and purchasing behaviour
- Evaluate product and regional performance
- Assess operational efficiency, especially delivery performance
- Provide actionable and implementable business recommendations

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4. Dataset Description

The dataset consists of structured transactional data with the following attributes:

Customer Attributes

- Customer ID
- Gender
- Age

Transaction Attributes

- Order ID
- Date of purchase
- Quantity purchased
- Price per unit

Product Attributes

- Product category

Operational Attributes

- Store type (Online/Offline)
- Delivery status
- Payment method

Geographical Attributes

- City

Derived Metric

- Revenue (calculated as price × quantity)

This dataset enables multi-dimensional analysis across customer, product, geographic, and operational dimensions.

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5. Data Cleaning & Preprocessing

Data preprocessing is a critical step to ensure the reliability and accuracy of the analysis. The following steps were performed:

- Converted date column into datetime format for time-series analysis
- Removed duplicate records to avoid bias

- Handled missing values to maintain dataset integrity
- Standardised column formats for consistency
- Created a new feature:
    - **Revenue = price_per_unit × quantity**
- Removed redundant columns such as total_amount and total_price

These steps ensured the dataset was clean, consistent, and suitable for analysis.

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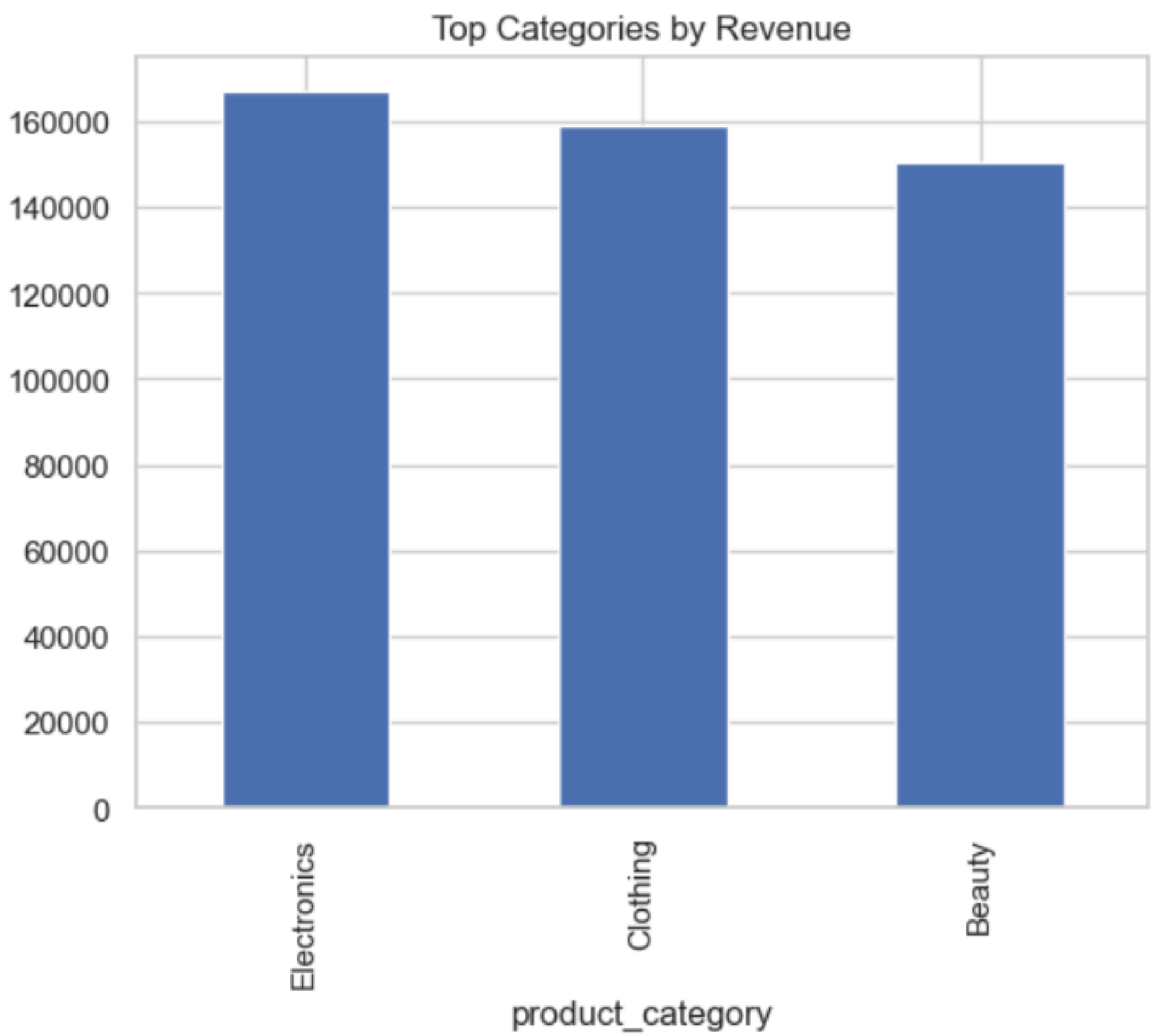
6. Exploratory Data Analysis (EDA)

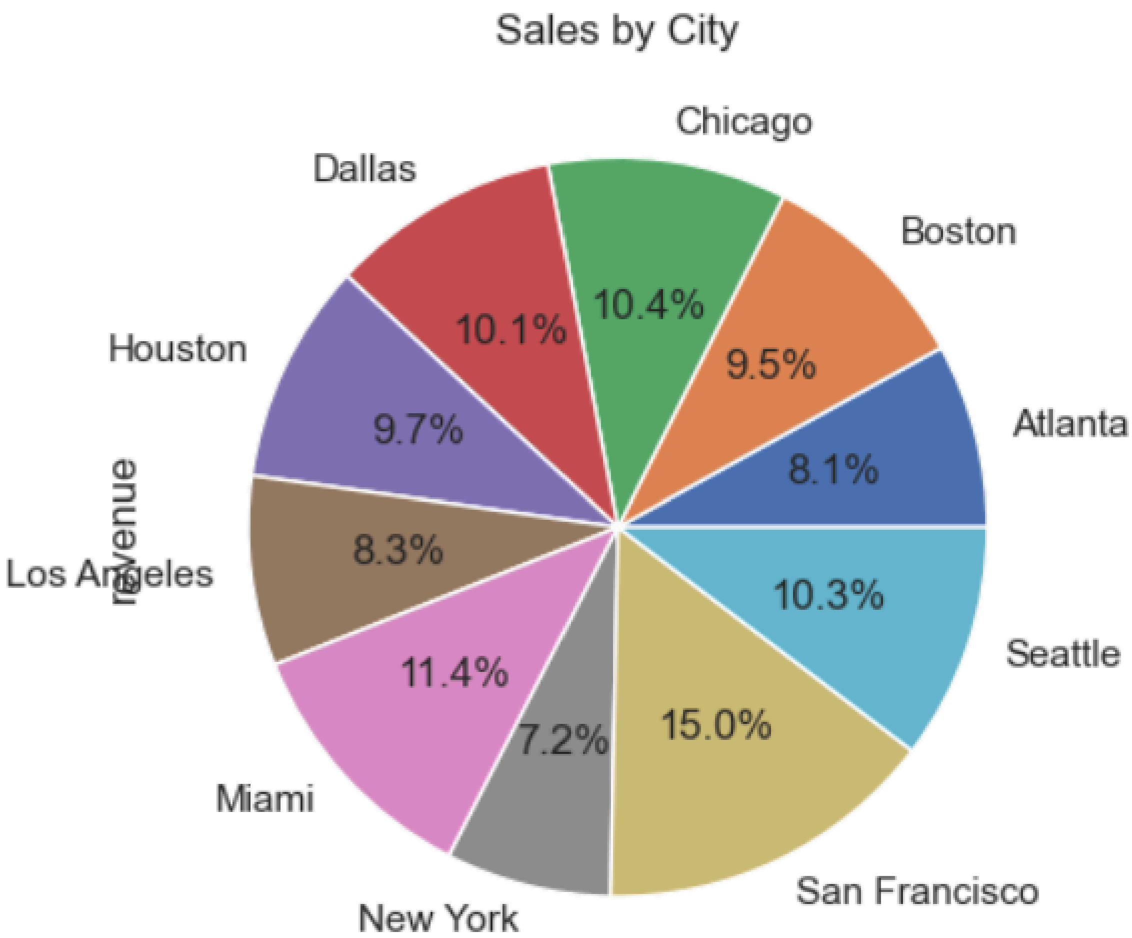
EDA was performed to understand data distribution, detect patterns, and generate initial insights.

Figure 1: Revenue by Product Category



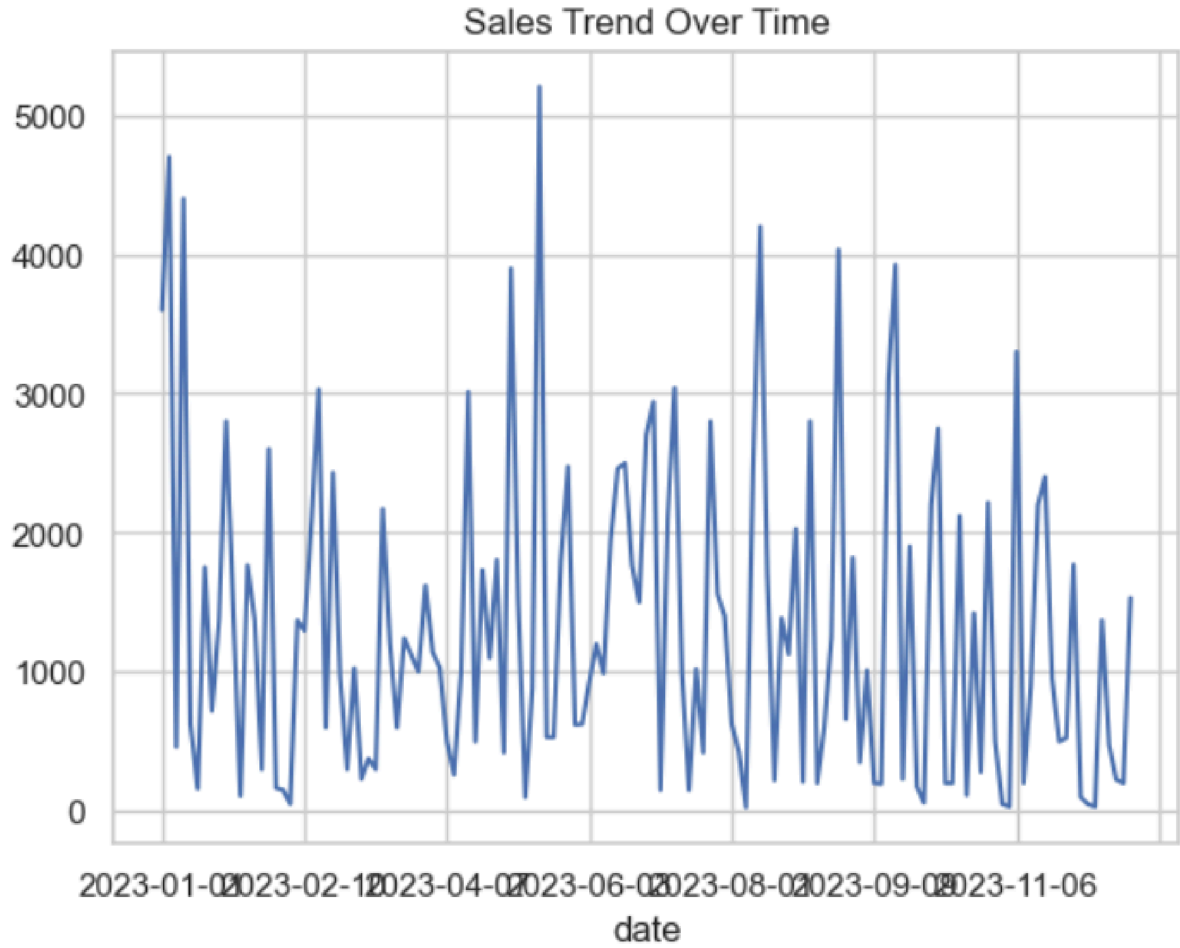
This visualisation shows that a small number of product categories contribute disproportionately to total revenue, indicating a Pareto-like distribution. This suggests that focusing on top-performing categories can significantly enhance profitability.

Figure 2: Top Cities by Revenue



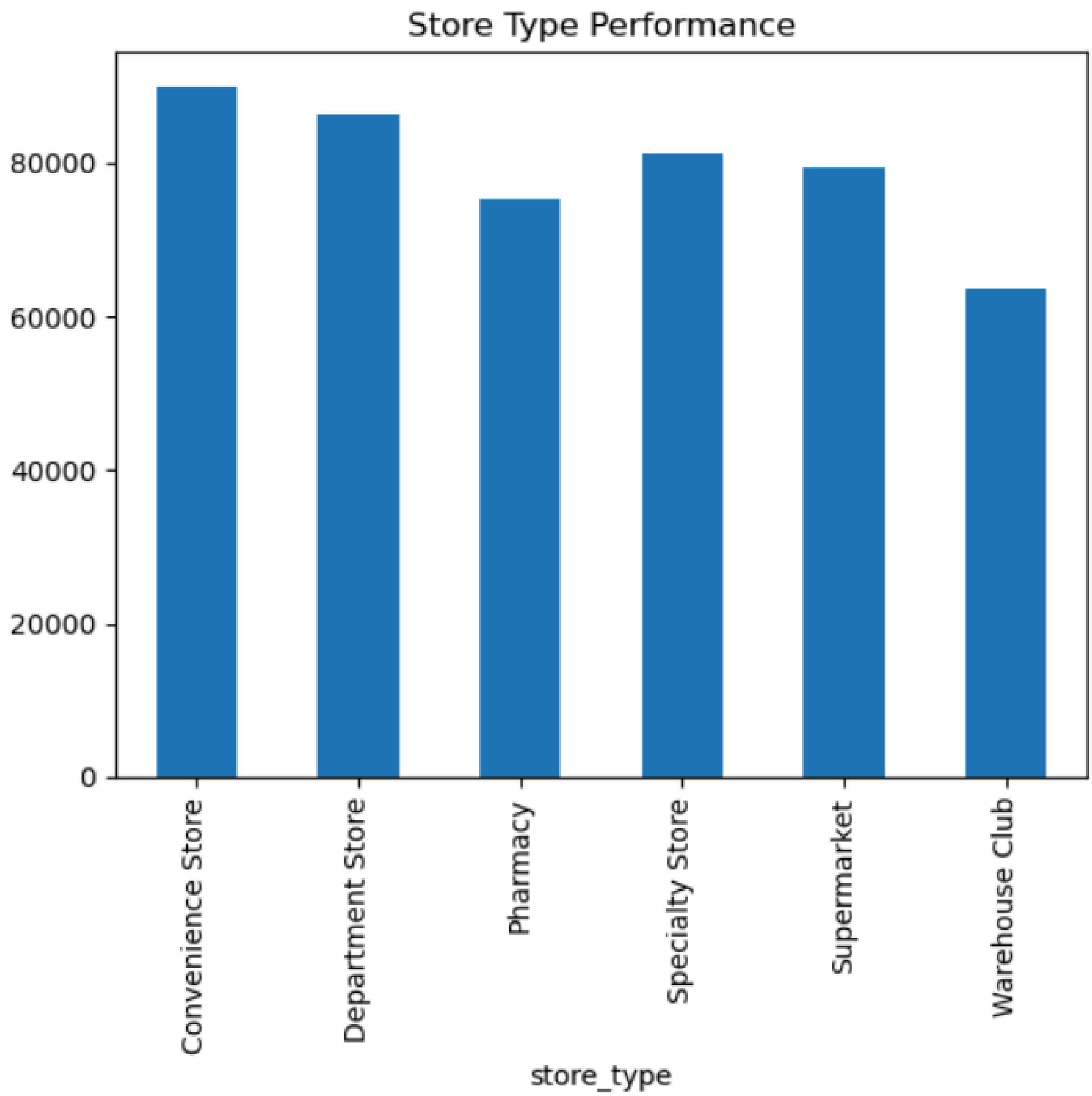
The analysis reveals geographic concentration of sales, where certain cities dominate revenue contribution. This insight can guide regional expansion and targeted marketing strategies.

Figure 3: Revenue Trend Over Time



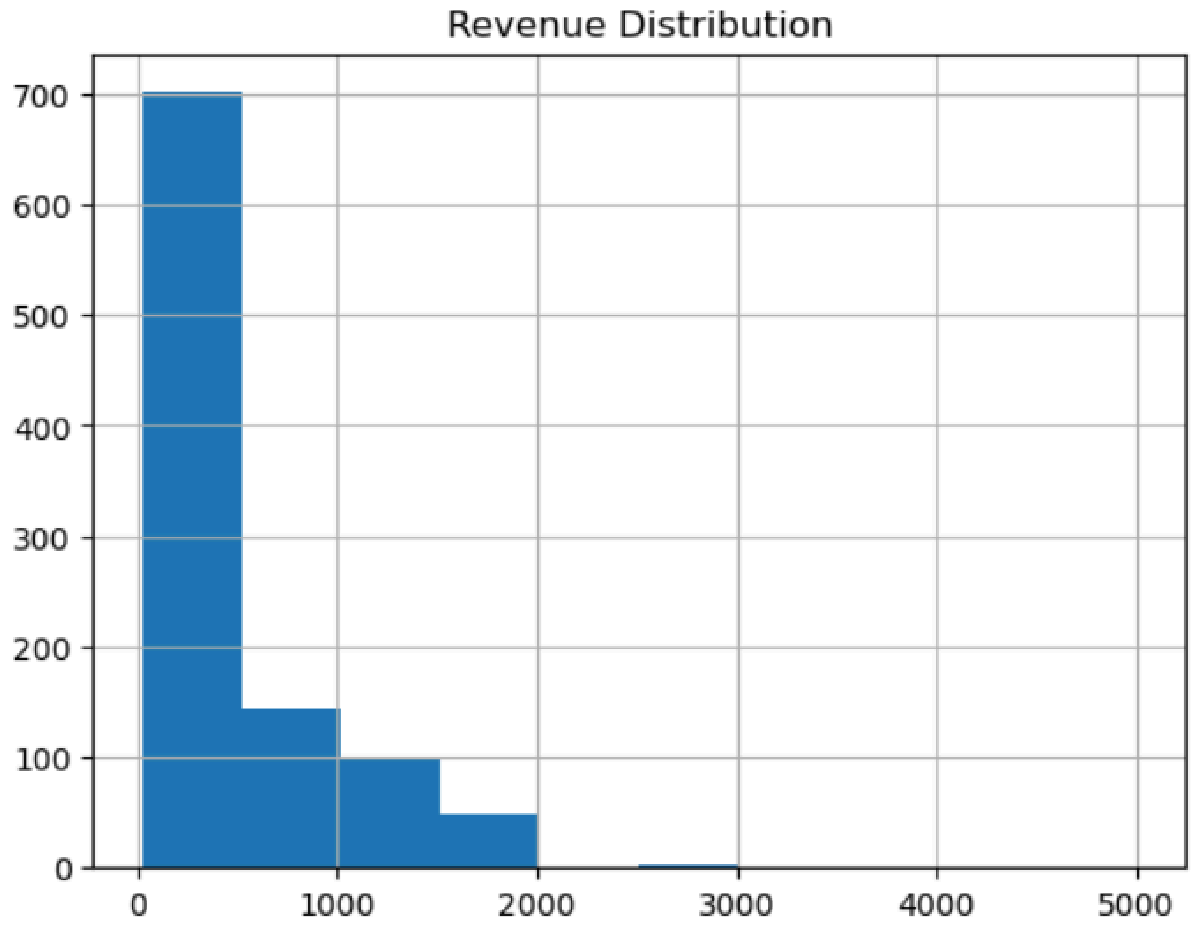
Time-series analysis highlights fluctuations in revenue, which may be influenced by seasonality, promotions, or external factors. Understanding these trends helps in forecasting and planning.

Figure 4: Store Type Performance



Comparison of store types provides insights into channel effectiveness. If online stores outperform offline stores, digital transformation strategies should be prioritised.

Figure 5: Revenue Distribution



The revenue distribution is right-skewed, indicating that while most transactions are small, a few high-value transactions contribute significantly to total revenue.

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7. Statistical & Advanced Analysis

7.1 Correlation Analysis

Correlation analysis helps identify relationships between numerical variables, such as:

- Price per unit
- Quantity
- Age
- Revenue

Key observations:

- Negative correlation between price and quantity suggests price sensitivity
- Positive correlation between quantity and revenue indicates volume-driven sales

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7.2 Customer Segmentation

Customers were segmented based on total revenue contribution:

- **High-value customers:** Above average revenue contribution
- **Low-value customers:** Below average contribution

This segmentation reveals that a small percentage of customers generates a large share of revenue, supporting the Pareto Principle (80/20 rule).

7.3 Hypothesis Testing

A t-test was conducted to evaluate differences in spending behaviour between male and female customers.

- **Null Hypothesis (H_0):** No difference in spending
- **Alternative Hypothesis (H_1):** A significant difference exists

This statistical approach ensures that conclusions are backed by evidence rather than assumptions.

7.4 Operational Analysis (Delivery Performance)

Delivery status analysis provides insights into operational efficiency:

- Successful deliveries contribute directly to revenue realisation
- Failed or delayed deliveries result in revenue loss and reduced customer satisfaction

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8. Key Insights

- Revenue is concentrated among a few product categories
- A small group of customers drives a large portion of sales
- Geographic regions show uneven performance
- Customer demographics influence purchasing patterns
- Delivery performance has a direct impact on revenue
- Store type plays a role in overall business success

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9. Business Recommendations

9.1 Product Strategy

- Focus on high-performing categories
- Optimise inventory for fast-moving products
- Reduce underperforming SKUs

9.2 Customer Strategy

- Implement loyalty and reward programs
- Personalise marketing campaigns
- Target high-value customer segments

9.3 Operational Strategy

- Improve delivery systems and logistics
- Monitor delivery KPIs
- Reduce failed delivery rates

9.4 Sales & Marketing Strategy

- Expand in high-performing cities
- Use data-driven pricing strategies
- Promote high-demand products

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10. Implementation Plan

Short-Term (0–3 Months)

- Identify top-performing products and customers
- Improve delivery tracking and efficiency
- Launch targeted marketing campaigns

Medium-Term (3–6 Months)

- Introduce customer loyalty programs
- Optimize pricing strategies
- Expand in high-performing regions

Long-Term (6–12 Months)

- Implement recommendation systems
- Invest in predictive analytics
- Expand business operations

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11. Limitations of the Study

- Limited historical data may affect trend analysis
- External factors (economic conditions, competition) are not included
- Assumptions made during data cleaning may impact results

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12. Future Scope

- Implement machine learning models for sales prediction
- Perform customer lifetime value (CLV) analysis
- Build dashboards using Power BI or Tableau
- Integrate real-time data for continuous monitoring

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13. Conclusion

This project demonstrates how retail data can be effectively used to generate actionable business insights. By leveraging statistical analysis and data visualisation techniques, the study provides a strong foundation for strategic decision-making.

The recommendations outlined in this report can significantly improve revenue growth, customer satisfaction, and operational efficiency, enabling the business to achieve long-term success.